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Performance analysis of fully depleted triple material surrounding gate (TMSG) SOI MOSFET

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Abstract In this paper, analytical model for threshold voltage is derived for fully depleted Triple material Surrounding gate (TMSG) SOI MOSFET. Three gate material of different work functions are introduced in the SOI MOSFET structure to reduce the short channel effects. The two dimensional Poisson equation is solved and based on parabolic approximation method, the model for threshold voltage is developed. The threshold voltage is analyzed for device parameters such as gate length ratios, oxide thickness, silicon thickness, doping concentration. The results of the analytical model values are validated using MEDICI simulation.

Keywords Triple material Surrounding Gate (TMSG) MOSFETs · Short channel effects (SCE) · Surface potential · Work function

1 Introduction

In the nanoscale regime, scaling of transistors has paved the way for reduction of cost and size of devices. When the classical MOS structures stands out in its inadequacy due to scaling and Short Channel effects, the devices so called Multigate MOSFETs were introduced and they found to be terrific in their performance. The ITRS recognizes the importance of these multigate MOSFETs and call them as "Advance Devices". While the dimensions of the transistors diminish, the close propinquity between the Source and the Drain reduces the capability of the gate electrode to direct the potential distribution and the flow of current in the channel region, and the adverse effects called Short Channel effects start plaguing MOSFETs. And the recent trend is to find out a device which has reduction in size as well as minimum of these so called short channel effects. This idea is a key to multigate MOSFET devices. Introducing more than one gate helps to reduce the short channel effects. In single gate structures, the gate length must be four times the thickness of the SOI layer to reduce the short channel effects. Whereas for multigate MOSFETs, the SCEs are reduced and the gate length can be two times the thickness of the SOI layer.

In recent years, a number of classical MOSFETs [1] have been proposed as the device structures to sustain the growth of the CMOS technology into nanoscale. These device structures include Double Gate (DG) MOSFETs, Surrounding Gate (SG) MOSFETs and FinFETs. The double gate MOSFETs consists of two gates, front gate and back gate to control the device. This structure has more advantages than single gate conventional MOSFETs. The short channel effects still dominate in these devices and hence in order to reduce the short channel effects the device is fully surrounded by gate material and this invents a device, surrounding gate MOSFET. The surrounding gate MOSFET [2–5] is one of the recent solutions to minimize shortchannel effects to the least.

To make the device more efficient, gate engineering was introduced. In the SOI MOSFET device, Long et al. [6] introduced gate engineering in MOSFETs and it produced a device, dual material gate MOSFETs (DMG). To augment the immunity against SCEs, a new structure, called

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a dual-material gate (DMG) MOSFET was anticipated [7]. This structure has two metals in the gate, each with different work functions which provides concurrent increase in transconductance and suppressed SCEs due to a step in the surface-potential profile when compared with a single-gate MOSFET. This causes an increased lifetime of the device, minimization of the ability of the localized charges to raise drain resistance, and more control of gate over the conductance of the channel so as to increase the gate transport efficiency. It was extended in DG MOSFETs and Reddy et al. introduced the Dual material Double gate (DMDG) MOSFETs. This structure inhibits the advantages of both DMG and DG MOSFETs. With this structure, improved transconductance and reduction in the electric field near the drain is reported [8]. The difficulty with DMDG MOSFETs is the control of the threshold voltage that is hardly dependent on the doping concentration and strongly affected by the thickness of silicon film. Kumar et al. [9] introduced gate engineering in surrounding gate MOSFET and he proposed dual material SG MOSFET (DMSG). It shows improved performance than DMDG structures. In order to make the device prominent with reduced short channel effects, a new structure is proposed in which gate is made up of three different materials [10] called as fully depleted Triple material Surrounding gate SOI MOSFET (TMSG MOSFET).We have already developed a model for surface potential and electric field [11] and subthreshold characteristics [12] of the device earlier. But still many device parameters has not been analyzed so far and hence in this paper, the device is studied with various device parameters.

2 Device structure

Figure 1 shows the schematic structure of the triple material surrounding gate MOSFET. In this structure, the gate electrode consists of three materials M1, M2 and M3 of different work functions deposited over respective lengths L_1, L_2 and L_3 on the gate oxide layers. The gate materials are chosen in such a way that $\phi_{M1} > \phi_{M2} > \phi_{M3}$ and $L = L_1 + L_2 + L_3$. The gate material at the source end is with the highest work function called the control gate. The middle material with the intermediate work function is called the first screen gate. The material at the drain end is with the lowest work function called as second screen gate. The device is composed of gate material 1 with the work function $\phi_{M1} = 4.8$ eV (Au), gate material 2 with $\phi_{M2} = 4.6$ eV (Mo), and gate material 3 with $\phi_{M3} = 4.4$ eV (Ti).

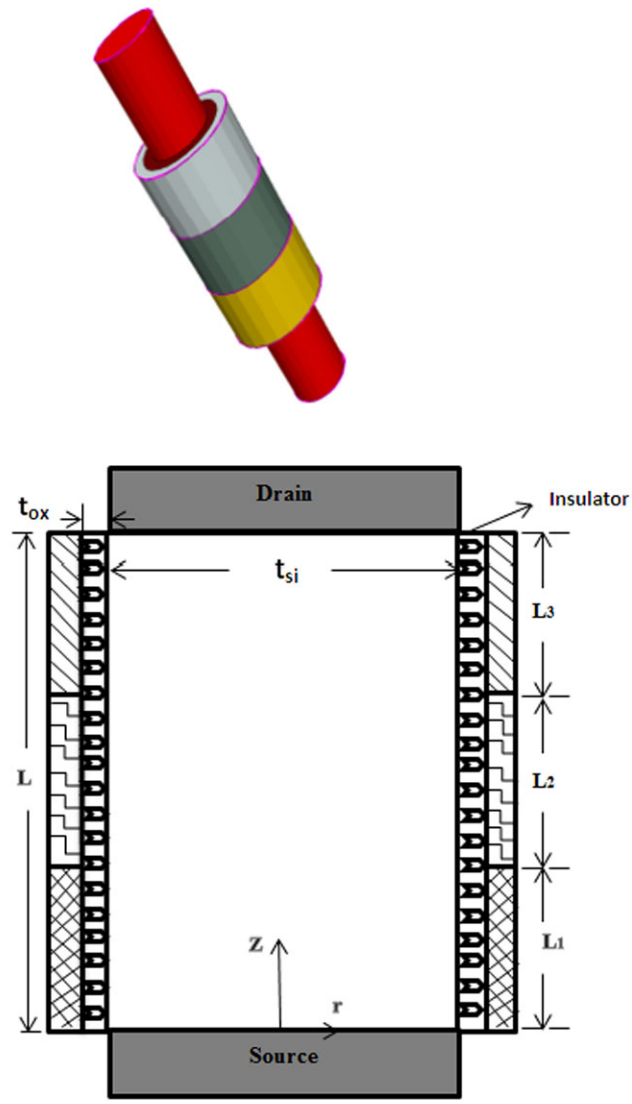

Fig. 1 Schematic cross sectional view of fully depleted triple material surrounding gate (TMSG) SOI MOSFET

3 Proposed model

In the literature, there are two general approaches to model the surface potential, electric field and threshold voltage. One such approach is the numerical simulation. The other approach is to develop an analytical solution. The analytical solution may be using either charge sharing approach or solving the Poisson equation. The solution to Poisson equation can be obtained by different methods in the literature. It includes Superposition method, Fourier series expansion, Green function [13], Newton Raphson method [14], Parabolic approximation method.

Chang et al. has proposed superposition method to solve the Poisson equation for DMG MOSFETs. For the same device, Saxena et al. used parabolic approximation method and made the solution easier and simple. Similarly for partially depleted DMG MOSFET, Kumar et al. used parabolic approximation method to solve the Poisson equation. Newton Raphson method was used by Syamal et al. [14] to find

Table 1 Expansion of symbols

Symbol	Description
ϕ_{M1}	Work function for material 1
ϕ_{M2}	Work function for material 2
ϕ_{M3}	Work function for material 3
L_1	Length of material 1
L_2	Length of material 2
L_3	Length of material 3
L	Device channel length
t_{ox}	Gate oxide thickness
t_{si}	Silicon thickness
q	charge
K_B	Boltzmann Constant
μ	Carrier mobility
N_A	Acceptor concentration
N_D	Doping concentration
ε_{ox}	Permittivity of silicon dioxide
ε_{si}	Permittivity of silicon
V_{gs}	Gate to source voltage
V_{ds}	Drain to source voltage

out the surface potential for Double gate DG MOSFETs. Green function can also be used to develop the surface potential model for nanoscale MOSFETs. Chang et al. has proposed solution for DMSG MOSFETs using superposition method [15]. Tiwari et al. has developed a surface potential model for TMDG MOSFETs using parabolic approximation method [16].

Till date, for TMSG MOSFETs, surface potential model was developed by Wang et al. using superposition method [10]. Since the method is slightly difficult and complex, in order to make the model easier, in the proposed work, the parabolic approximation method [11, 12] has been adopted to develop a model for threshold voltage. Moreover, the parabolic approximation method has a uniform profile throughout the approximation.

The coordinate system consists of a radial direction r, a vertical direction z and an angular component θ in the plane of the radial direction, as shown in Fig. 1. Assuming the impurity density in the channel region to be uniform, and neglecting the effect of the fixed oxide charges on the electrostatics of the channel, the potential distribution in the silicon thin film before the onset of strong inversion can be written as

$$\frac{\partial^2}{\partial r^2}\phi_i(r,z) + \frac{1}{r}\frac{\partial}{\partial r}\phi_i(r,z) + \frac{\partial^2}{\partial z^2}\phi_i(r,z) = \frac{qN_A}{\varepsilon_{si}}$$
$$i = 1, 2, 3 \tag{1}$$

3.1 Boundary conditions

The Poisson's equation is solved separately under the three regions using the following boundary conditions.

(a) The surface potentials at the interfaces of the three dissimilar metals are continuous:

$$\begin{aligned}\phi_1(r, L_1) &= \phi_2(r, L_1),\\ \phi_2(r, L_1 + L_2) &= \phi_3(r, L_1 + L_2).\end{aligned} \tag{2}$$

(b) The electric field at the interfaces of the three dissimilar metals are continuous:

$$\begin{aligned}\left.\frac{\partial\phi_1(r,z)}{\partial z}\right|_{z=L_1} &= \left.\frac{\partial\phi_2(r,z)}{\partial z}\right|_{z=L_1},\\ \left.\frac{\partial\phi_2(r,z)}{\partial z}\right|_{z=L_1+L_2} &= \left.\frac{\partial\phi_3(r,z)}{\partial z}\right|_{z=L_1+L_2}.\end{aligned} \tag{3}$$

(c) The electric flux at the gate/oxide interface is continuous for both metal gates:

$$\left.\frac{\partial\phi_i(r,z)}{\partial r}\right|_{r=t_{si}/2} = \frac{\varepsilon_{ox}}{\varepsilon_{si}} \times \frac{V_{gs} - V_{FBi} - \phi_i(r,z)}{t1}$$
$$\text{where } i = 1, 2, 3 \tag{4}$$

Where

$$t1 = r \times \ln\left(1 + \frac{2t_{ox}}{t_{si}}\right). \tag{5}$$

(d) The potential at the source end is

$$\phi_1(r, 0) = V_{bi}. \tag{6}$$

Where built in potential is given by

$$V_{bi} = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right) \tag{7}$$

Where $V_T = k_B T/q$ (k_B is the Boltzmann Constant and T is the temperature).

(e) The potential at the drain end is

$$\phi_i(r, L_1 + L_2 + L_3) = V_{bi} + V_{ds}. \tag{8}$$

The potential profile in the vertical direction, i.e., the z-dependence of $\phi(r,z)$ can be approximated by a simple parabolic function as proposed by for triple material surrounding gate SOI MOSFET as

$$\phi(r,z) = c_1(z) + c_2(z)r + c_3(z)r^2 \tag{9}$$

Substituting the boundary conditions (1)–(5) in Eq. (9) we obtain

$$\frac{d^2\phi_{s1}(z)}{dz^2} - K\phi_{s1}(z) = M_1 \quad 0 \le z \le L_1 \tag{10}$$

$$\frac{d^2\phi_{s2}(z)}{dz^2} - K\phi_{s2}(z) = M_2 \quad L_1 \le z \le L_1 + L_2 \tag{11}$$

$$\frac{d^2\phi_{s3}(z)}{dz^2} - K\phi_{s3}(z) = M_3 \quad L_1 + L_2 \le z \le L_3 \tag{12}$$

Where

$$K = \frac{2\varepsilon_{ox}}{R^2\varepsilon_{si}(\ln(1+\frac{t_{ox}}{R}))} \tag{13}$$

$$\begin{aligned} M_1 &= \frac{qN_A}{\varepsilon_{si}} - K^2(V_{gs} - V_{FB1}); \\ M_2 &= \frac{qN_A}{\varepsilon_{si}} - K^2(V_{gs} - V_{FB2}) \\ M_3 &= \frac{qN_A}{\varepsilon_{si}} - K^2(V_{gs} - V_{FB3}); \quad R = t_{si}/2 \end{aligned} \tag{14}$$

Flat band voltages are given as

$$V_{FBn} = \phi_{Mn} - \phi_{si}, \quad n = 1, 2, 3. \tag{15}$$

ϕ_{si} is the silicon work function which is given by

$$\phi_{si} = \chi_{si} + \frac{E_g}{2q} + \phi_F \tag{16}$$

where $\phi_F = V_T \ln(\frac{N_A}{n_i})$.

Solving the second order differential equation using particular Integral and Complementary function

$$\phi_{S1}(z) = A_1e^{Kz} + B_1e^{-Kz} - \frac{M_1}{K^2} \quad 0 \le z \le L_1 \tag{17}$$

$$\phi_{S2}(z) = A_2e^{Kz} + B_2e^{-Kz} - \frac{M_2}{K^2} \quad L_1 \le z \le L_1 + L_2 \tag{18}$$

$$\phi_{S3}(z) = A_3e^{Kz} + B_3e^{-Kz} - \frac{M_3}{K^2} \quad L_1 + L_2 \le z \le L_3 \tag{19}$$

Substituting the boundary conditions in the above equations, the values for various constants are given as

$$B_3 = \frac{2\delta V_{DS} + 2\delta D_2 + (\delta\delta_3 + \delta^2\delta_1\delta_2)D_3 - \delta\delta_2\delta_3 D_1}{\delta_2(2 - 2\delta^2 + \delta_2^2\delta_3^2)}$$

$$B_1 = \left(\delta_1\left(Q_1 - B_3\delta_1^{-1}\delta_3^{-1}\right) + \left(\frac{\delta_1\delta_3}{2}\right)D_3 + \left(\frac{\delta\delta_2}{2}\right)D_3 + B_3\delta_2^2\delta_3 - V_{bi}\delta_1\delta - Q_2\delta_1\delta - \delta D_1\right)\Big/\left(\delta\left(\delta_1^{-1} - \delta_1\right)\right)$$

$$B_2 = \frac{B_3\delta_2\delta_3 + (\frac{\delta}{2})D_3}{\delta_1\delta_3}$$

$$A_2 = \frac{\delta_1(Q_1 - B_3\delta_1^{-1}\delta_3^{-1}) + (\frac{\delta_1\delta_3}{2})D_3}{\delta}$$

$$A_3 = \frac{Q_1 - B_3\delta_1^{-1}\delta_3^{-1}}{\delta_1\delta_3}$$

$$A_1 = V_{bi} + Q_2 - B_1$$

Where

$$\begin{aligned} &\delta = e^{KL} = e^{K(L_1+L_2+L_3)}, \quad \delta_1 = e^{KL_1}, \\ &\delta_1^{-1} = e^{-KL_1}, \quad \delta_2 = e^{KL_2} \\ &\delta_2^{-1} = e^{-KL_2}, \quad \delta_3 = e^{KL_3}, \quad \delta_3^{-1} = e^{-KL_3} \end{aligned} \tag{20}$$

$$\begin{aligned} &D_1 = \frac{M_2 - M_1}{K^2}, \quad D_2 = \frac{M_3 - M_1}{K^2}, \\ &D_3 = \frac{M_2 - M_3}{K^2} \end{aligned} \tag{21}$$

$$Q_1 = V_{bi} + V_{ds} + \frac{M_3}{K^2}, \quad Q_2 = M_1/K^2 \tag{22}$$

The electric field distribution along the channel length can be obtained by differentiating the surface potential given by (17), (18) and (19). As the electron transport velocity through the channel is related to the electric field pattern along the channel, the electric field component, in the z direction under M_1 is obtained as

$$\begin{aligned} E_1(z) &= \left.\frac{d\phi_{S1}(r,z)}{dz}\right|_{r=R} \\ &= A_1Ke^{Kz} - B_1Ke^{-Kz} 0 \le z \le L_1 \end{aligned} \tag{23}$$

$$\begin{aligned} &E_2(z) = \left.\frac{d\phi_{S2}(r,z)}{dz}\right|_{r=R} = A_2Ke^{K(z-L_1)} - B_2Ke^{-K(z-L_1)} \\ &L_1 \le z \le L_1 + L_2 \end{aligned} \tag{24}$$

$$\begin{aligned} E_3(z) &= \left.\frac{d\phi_{S3}(r,z)}{dz}\right|_{r=R} = A_3Ke^{K(z-(L_1+L_2))} \\ &\quad - B_3Ke^{-K(z-(L_1+L_2))} \quad L_1 + L_2 \le z \le L_3 \end{aligned} \tag{25}$$

In the case of the TMSG structure, due to the coexistence of three metal gates with different work functions, the surface potential minimum is solely determined by the metal gate with the higher work function. Therefore the minimum surface potential of the silicon pillar under the gate can be calculated as, differentiating $\phi_{S1}(z)$ with respect to z and equating to zero.

$$\left.\frac{d\phi_{S1}(z)}{dz}\right|_{z=z_{\min}} = 0 \tag{26}$$

$$\phi_S(Z_{\min}) = 2\sqrt{AB} - \left(M/K^2\right) \tag{27}$$

Where $z_{1\min} = \frac{1}{2K}\ln(\frac{B_1}{A_1})$.

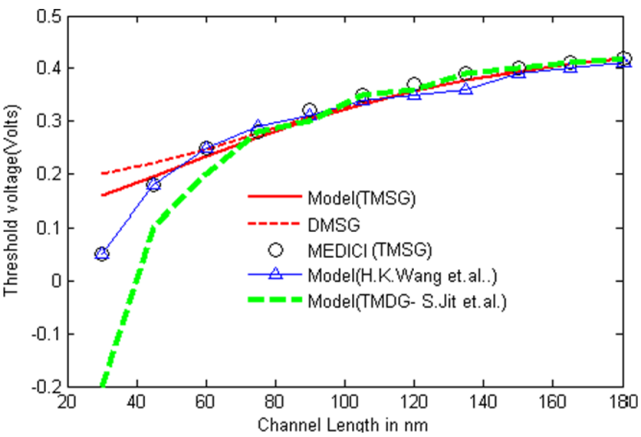

Fig. 2 Variation of threshold voltage for TMSG and DMSG MOSFET-comparison with other models

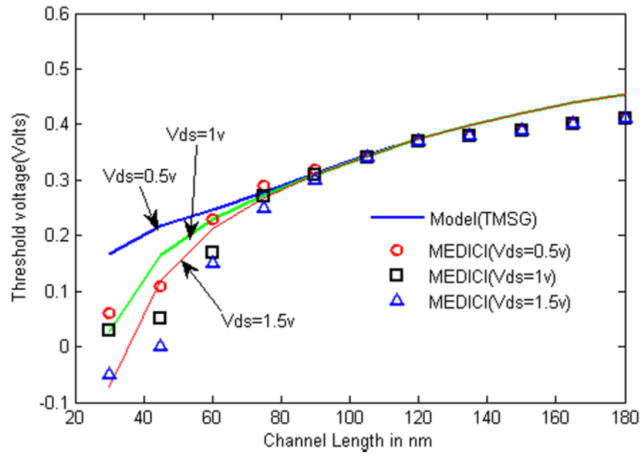

Fig. 3 Variation of threshold voltage of TMSG MOSFET for various drain bias $V_{ds} = 0.5$ V, 1 V and 1.5 V

The threshold voltage is defined as the gate voltage in region 1 at which the minimum surface potential is twice the bulk potential,

$$\phi_{S1,\min} = 2\phi_B$$

By substituting $V_{gs} = V_t$ in Eq. (27) and solving for V_t, the threshold voltage is obtained as

$$V_t = \frac{qN_a}{\varepsilon_{si}K^2} - 2\sqrt{A_1B_1}K^2 + 2\phi_B K^2 + V_{FB1} \tag{28}$$

4 Results and discussions

The threshold voltage is calculated and plotted for various device parameters. The analytical model is validated with the results obtained from numerical simulation tool TCAD-MEDICI and it well agrees with the obtained values.

Simulation Parameters used in the proposed structure

$t_{ox} = 2$ nm, 2R = 40 nm, $V_{gs} = 0.1$ V,

$N_A = 10^{16}$, $V_{ds} = 0.5$ V, $\phi_{M1} = 4.8$ eV (Au),

$\phi_{M2} = 4.6$ eV (Mo), $\phi_{M3} = 4.4$ eV (Ti).

In Fig. 2 the threshold voltage is plotted for different channel lengths. The figure compares the threshold voltage for TMSG and DMSG structure of MOSFETs. It is seen from the figure that threshold voltage decreases when the channel length decreases. From the result, the threshold voltage "roll-off" is observed for decreased channel lengths. This results in reduced short channel effects. This is because as the channel length is scaled down, the gate loses its control and drain bias starts to initiate DIBL slowly. The model is compared with superposition model proposed by Wang et al. It is also compared with Triple material double gate (TMDG) MOSFETs.

Figure 3 shows variation of threshold voltage in Y axis for channel length along X axis. It is plotted for various drain bias. It is evident from the figure that when the channel length decreases, it induces DIBL specifically due to which there is a variation in the threshold voltage. The increase in drain to source voltage causes decrease in threshold voltage. When the drain to source voltage is increased, due to DIBL the threshold voltage is lowered. Compared to TMDG MOSFETs and DMSG MOSFETs, the threshold voltage is improved and it shows better performance.

The threshold voltage is plotted across the channel length for different L1:L2:L3 ratios. L1, L2 and L3 represents the lengths of the gate material with different work function. From the Fig. 4, it is observed that higher gate length ratios have higher threshold voltages which makes is not suitable for low power applications. Whereas lower gate length ratios have lesser threshold voltage. Further, high gate ratio effectively decreases the threshold roll off when the channel length is decreased. The gate length ratio varies the proportion of the gate material 1, 2 and 3. This simultaneously varies the workfunction difference which varies the threshold voltage. When the ratio increase, it increases the gate material with the higher work function. Moreover, high gate length ratios also increase the power dissipation. It is clear from the fact that the device should have lesser gate length ratios to make it suitable for nanoscale applications.

The threshold voltage is plotted with channel length along the X axis for various oxide thickness in Fig. 5. It is observed that the threshold voltage is inversely proportional to the oxide thickness. When the oxide thickness $t_{ox} = 1$ nm, the threshold voltage is high and it also produces more short channel effects. From the graph, it can be observed that the threshold voltage increases with decreased oxide thickness. When the thickness of the oxide layer is less, it introduces

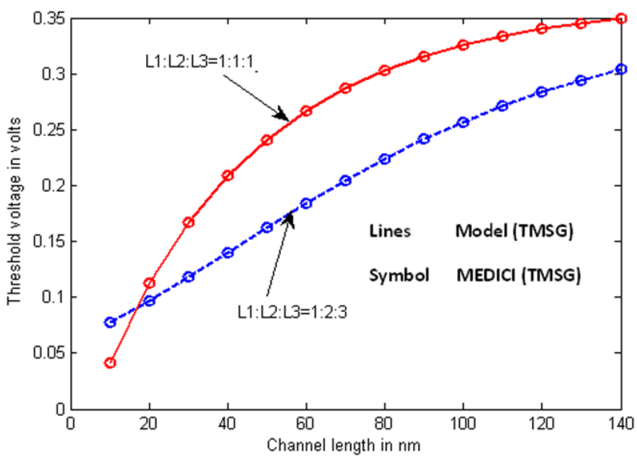

Fig. 4 Threshold voltage for different gate length ratios

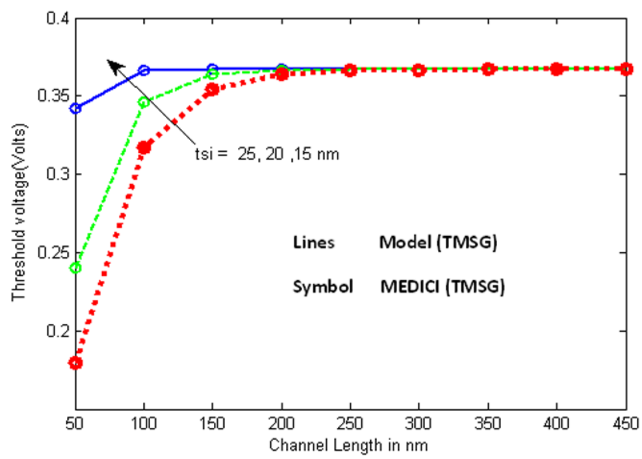

Fig. 6 Threshold voltage plotted with channel length for different silicon thickness

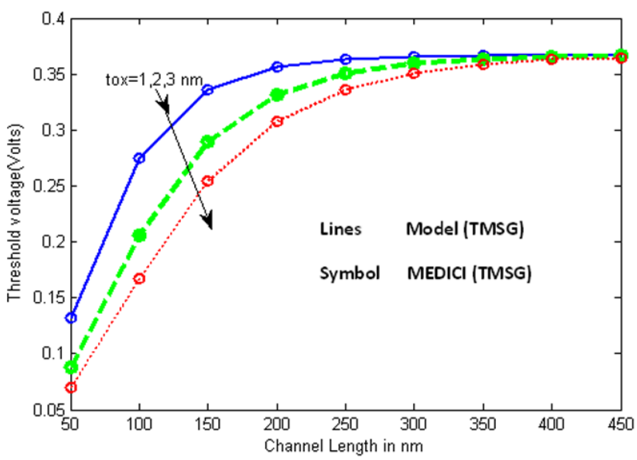

Fig. 5 Variation of threshold voltage for various oxide thickness

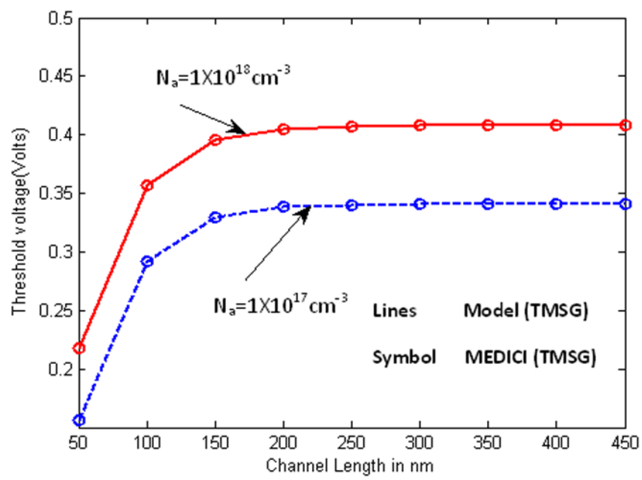

Fig. 7 Variation of threshold voltage for different doping profile

some leakage currents in the subthreshold condition because due to less insulating layer, the electrons and charge carriers may tunnel through the device to produce leakage currents called as subthreshold current. It is better to have some nominal value of oxide thickness in order to have less short channel effects as well as to reduce the leakage current.

The variation of threshold voltage with channel length for various diameter of silicon is plotted in Fig. 6. It is shown from the result that the threshold voltage is high for less value of silicon thickness. The threshold voltage varies inversely with the silicon thickness. This is obtained due to the improvement in gate electrostatics on the channel for decreasing silicon thickness. It is obvious that high threshold voltage also produces more short channel effects and it makes the device not suitable for low power applications.

The threshold voltage is plotted with channel length for various doping concentration in Fig. 7. It is evident that as the doping concentration increases, there is a roll-up in the threshold voltage. By varying the doping concentration, the barrier height is increased. Hence the voltage required to deplete the charge carriers is also increased. This makes the transition voltage to be high that increases the threshold voltage.

The threshold voltage is plotted for various built in potential in Fig. 8. The built in potential is varied and the values of threshold voltage are plotted. The threshold voltage varies for built in potential and it saturates at a certain voltage beyond which it remains constant. This is achieved for low values of potential when the gate length is reduced. Usually, the shift of minimum surface potential towards the source side increases when the built in potential increases and hence the DIBL increases. But in the proposed TMG structure, the shift in the position is zero due to the gate engineered structure and hence DIBL is comparatively reduced. This proves the reduction of short channel effects. The analytical modeling results are validated by MEDICI simulation results.

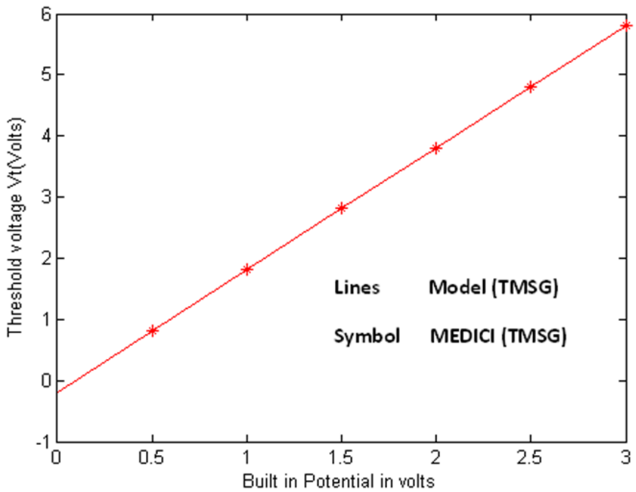

Fig. 8 Threshold voltage variation for different built in potential

5 Conclusion

In this paper, the threshold voltage model of a fully depleted Triple material surrounding gate MOSFET is developed by solving the Poisson equation. The results reveal the importance of gate engineering in SOI MOSFETs. Because the results compare the dual material gate MOSFETs with the proposed triple material structure, proving to be advantageous than the dual structures. Since the work function difference at the interface of the dissimilar metals produces more advantages, it is three interfaces formed in the proposed model, responsible for reducing the SCEs. The threshold voltage is analyzed for various device parameters like silicon thickness, oxide thickness, gate length ratios, doping concentration. The immunity offered against short channel effects and the reduction of SCEs are discussed and studied. The analytical model results are validated by MEDICI simulation results.

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